

SHOCK EVOLUTION IN SUPERNOVA 1987A: TWO DECADES OF CHANDRA OBSERVATIONS

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ABSTRACT

This study presents a detailed investigation of the long-term evolution of soft (0.5-2 keV) and hard (3-8 keV) X-ray fluxes from the supernova remnant SN 1987A, based on multi-epoch observations obtained from the Chandra X-ray Observatory. The primary aim is to examine the dynamical interaction between the expanding shock front and the surrounding circumstellar medium (CSM), and to characterise the transition from a ring-dominated emission phase to an ejecta-dominated regime.

The result indicates that the soft X-ray flux exhibits a pronounced increase during the early stages, corresponding to the interaction of the forward shock with the dense equatorial ring, followed by a progressive flattening and subsequent decline, signifying the passage of the shock beyond this structure. In contrast, the hard X-ray flux demonstrates a sustained and systematic increase, indicative of emission from higher temperature plasma components and an increasing contribution from reverse-shocked ejecta.

The temporal evolution of the hardness ratio provides robust evidence for a transition in the dominant emission mechanisms. A positive trend in the late-time hardness ratio suggests an enhanced contribution from high-energy processes, although the moderate correlation reflects the inherent complexity of the shock-CSM interaction and the inhomogeneous nature of the ambient medium.

Overall, the findings establish that SN 1987A has entered a subsequent evolutionary phase in which the X-ray emission is governed by interactions with lower-density circumstellar material and the emergence of reverse-shocked ejecta. This work provides important observational constraints on shock dynamics, plasma emission mechanisms, and the long-term evolution of young supernova remnants.

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CHAPTER - 1

INTRODUCTION

1.1 Overview of Supernovae

Supernovae are among the most energetic astrophysical events in the universe, marking the violent end stages of stellar evolution. These explosions release an enormous amount of energy ($\sim 10^{51}$ erg), enriching the interstellar medium with heavy elements and driving shock waves that influence star formation processes. Supernovae are broadly classified into Type-I and Type-II based on their spectral features and progenitor systems. Core-collapse supernovae, such as Type-II events, originate from massive stars ($M > 8 M_{\odot}$) that undergo gravitational collapse after exhausting their nuclear fuel.

The remnants of supernovae, known as supernova remnants (SNRs), evolve as the ejected material interacts with the surrounding medium. This interaction produces radiation across the electromagnetic spectrum, particularly in X-rays, where high-energy processes such as shock heating dominate.

1.2 SN 1987A : Historical significance

Supernova 1987A, discovered on February 23, 1987, in the Large Magellanic Cloud (LMC), represents the closest observed supernova in modern times, at a distance of approximately 51 Kpc. Its proximity has enabled unprecedented multi-wavelength observations over several decades, making it a cornerstone object in astrophysical research.

The progenitor star, Sanduleak -69° 202, was identified as a blue supergiant, which was unexpected based on prior theoretical models that predicted red supergiant progenitors for Type-II supernovae. This discrepancy provided new insights into stellar evolution, particularly regarding mass loss and the late stages of massive stars.

Over the past decades, SN 1987A has been observed using various space-based observatories, including the Chandra X-ray Observatory, which has provided high-resolution imaging and spectroscopy. These observations have made SN 1987A a unique astrophysical laboratory for studying the early evolution of supernova remnants.

1.3 Structure of SN 1987A (Triple Ring System)

One of the defining features of SN 1987A is its complex circumstellar environment, characterized by a system of three rings: a dense inner equatorial ring and two larger outer rings. These structures

are believed to have formed from interactions between stellar winds during different evolutionary phases of the progenitor star.

The equatorial ring, with a diameter of approximately 1.4 light-years, is composed of dense clumps of circumstellar material. This ring plays a critical role in the evolution of the remnant, as it interacts with the outward-propagating shock wave. The outer rings are less dense and are not the primary contributors to the observed X-ray emission at present.

As the blast wave expands, it eventually encounters the dense equatorial ring, resulting in strong shock heating and enhanced X-ray emission. The geometry and density distribution of these structures provide valuable information about the mass-loss history of the progenitor star.

1.4 X-ray Emission Mechanisms

X-ray emission in supernova remnants arises from the interaction between the expanding shock front and the surrounding medium. When the shock wave propagates through the circumstellar medium, it heats the gas to temperatures of the order of 10^6 – 10^7 K, producing thermal X-ray emission.

The dominant emission mechanisms include thermal bremsstrahlung and line emission from highly ionized elements such as oxygen, neon, silicon, and iron. The X-ray flux depends strongly on the density of the emitting plasma and can be expressed as

$$F_x \propto n_e^2 V$$

where n_e is the electron density, and V is the emitting volume. This quadratic dependence implies that regions of higher density produce significantly stronger X-ray emission.

In addition to the forward shock that propagates into the circumstellar medium, a reverse shock travels inward through the ejecta. This reverse shock heats the inner material, contributing to higher-energy (hard X-ray) emission at later stages of evolution.

1.4.1 Evolutionary Phase of SN 1987A

The temporal evolution of SN 1987A can be understood in terms of distinct interaction phases:

- Early Phase (~ 0 - 17 years): The shock propagates through a relatively low-density H II region, resulting in weak X-ray emission.
- Ring-Dominated Phase (~ 17 -24 Years): The shock interacts with the dense equatorial ring, leading to a rapid increase in soft X-ray emission due to efficient heating of dense clumps.

- Post-Ring / Ejecta-Dominated Phase (>24 years): The shock moves beyond the equatorial ring and encounters lower-density material. During this phase, the soft X-ray flux stabilizes or declines, while hard X-ray emission increases due to contributions from higher temperature plasma and reverse-shocked ejecta.

These phases provide a framework for interpreting the observed evolution of X-ray flux and spectral properties.

1.5 Objectives of the study

The objectives of this study are as follows:

- To analyze the temporal evolution of soft (0.5-2 keV) and hard (3-8 keV) X-ray fluxes of SN 1987A using multi-epoch Chandra observations.
- To evaluate the variation of the hardness ratio and its significance in understanding the underlying emission mechanisms.
- To investigate the interaction between the expanding shock wave and the circumstellar medium, particularly the equatorial ring.
- To identify and characterize the transition from the ring-dominated phase to an ejecta-dominated phase.
- To provide a physical interpretation of the observed trends in terms of shock dynamics, plasma emission processes, and density variations.

This study aims to contribute to a comprehensive understanding of the long-term evolution of supernova remnants and the physical processes governing their X-ray emission.

CHAPTER - 2

LITERATURE REVIEW

2.1 Early X-ray Observations

Initial X-ray observations of SN 1987A revealed relatively weak emission during the first decade following the explosion[1]. This behaviour was attributed to the propagation of the shock wave through a low-density H II region surrounding the progenitor[7]. Early studies indicated that the X-ray emission was dominated by thermal processes associated with shock-heated plasma, with temperatures of the order of 10^6 K[8].

With the advent of high-resolution instruments such as the Chandra X-ray Observatory[10], it became possible to resolve the remnant spatially and spectrally. These observations provided the first detailed insights into the structure and evolution of the X-ray-emitting regions, establishing a strong connection between shock dynamics and observed emission[2].

2.2 Evolution of X-ray light curves

Long-term monitoring of SN 1987A has enabled the construction of X-ray light curves across different energy bands[1],[4]. Observational studies have shown a steady increase in soft X-ray flux as the shock approached and interacted with the dense equatorial ring[3]. This phase was characterized by enhanced emission due to efficient shock heating of high-density material[2].

As the shock progressed through the ring, the soft X-ray flux reached a peak and subsequently exhibited a gradual flattening[4]. In contrast, the hard X-ray flux displayed a more gradual and continuous increase, indicating the presence of higher temperature plasma components and evolving shock conditions[6].

These observations highlight the importance of multi-band analysis in distinguishing between different physical regimes governing the emission[5].

2.3 Shock Interaction with Circumstellar Medium

The interaction of the supernova shock wave with the circumstellar medium (CSM) is the primary driver of X-ray emission in SN 1987A[7]. The dense equatorial ring represents a region of enhanced density, leading to strong shock heating and increased X-ray luminosity[3].

Studies have shown that X-ray flux is proportional to the square of the plasma density, implying

that dense clumps within the ring significantly contribute to the observed emission[8],[9].

Studies have shown that X-ray flux is proportional to the square of the plasma density, implying that dense clumps within the ring significantly contribute to the observed emission[8],[9]. The interaction between the shock and these clumps produces localized hotspots and complex emission structures[2].

As the shock exits the equatorial ring, it encounters lower-density material, resulting in changes in both flux intensity and spectral characteristics[5]. This transition marks a significant stage in the evolution of the remnant[6].

2.4 Recent Developments (Chandra Observations)

Recent Chandra observations indicate that SN 1987A is transitioning from a ring-dominated phase to a more advanced evolutionary stage[6]. The soft X-ray flux has shown signs of levelling off and a gradual decline, suggesting that the forward shock has propagated beyond the densest regions of the equatorial ring[5].

At the same time, the hard X-ray emission continues to increase, reflecting contributions from higher temperature plasma and reverse-shocked ejecta[8]. The evolution of the hardness ratio further supports this transition, indicating an increasing dominance of high-energy processes[6].

These findings are consistent with theoretical models of supernova remnant evolution, where the emission shifts from density-dominated regimes to temperature-dominated regimes as the shock interacts with lower-density environments[7].

2.5 Summary

The reviewed literature demonstrates that the interaction between the expanding shock wave and the surrounding circumstellar medium governs the X-ray evolution of SN 1987A. The transition from a ring-dominated phase to an ejecta-dominated phase represents a key stage in the remnant's evolution.

Previous studies emphasize the importance of long-term observations, spectral analysis, and flux evolution in understanding these processes. Building on these findings, the present work focuses on a detailed analysis of soft and hard X-ray flux evolution and their physical interpretation.

Table 2.1: Summary of previous studies on SN 1987A

Author & Year	Instrument/Data	Period	Key Findings	Relevance
Park et al. (2004)	Chandra ACIS	2000–2004	Flux increased as shock reached ring	Early soft X-ray dominance
Dewey et al. (2012)	Chandra HETG	2007–2011	Two shock components identified	Multi-temperature plasma
Helder et al. (2013)	Chandra + Optical	2010–2013	Shock interaction with dense clumps	Shock–emission relation
Frank et al. (2016)	Chandra + XMM	2000–2015	Flux peak (~2013), then decline	Transition phase
Alp et al. (2021)	Chandra monitoring	2015–2020	Decline in soft X-ray	Cooling + compact object
Ravi et al. (2024)	Chandra long-term monitoring	2000-2024	Decline in soft X-ray flux and continued increase in hard X-ray emission	Confirms transition to ejecta-dominated phase

CHAPTER - 3

DATA AND METHODOLOGY

3.1 Chandra X-ray Observatory Overview

The Chandra X-ray Observatory is one of NASA's flagship space telescopes, designed for high-resolution X-ray imaging and spectroscopy[10]. It operates in the energy range of approximately 0.1-10 keV and provides sub-arcsecond angular resolution, making it particularly suitable for studying compact and evolving astrophysical objects such as supernova remnants.

Chandra is equipped with advanced instruments, including the Advanced CCD imaging spectrometer (ACIS), which enables both imaging and spectral analysis[10]. The high spatial and spectral resolution of Chandra allows for a detailed investigation of shock structures, plasma conditions, and temporal evolution in supernova remnants like SN 1987A.

3.2 Description of Observation Data (ObsIDs)

This study uses multi-epoch Chandra X-ray Observatory observations of SN 1987A. Each observation is identified by a unique observation ID (ObsID) that corresponds to specific observation dates and exposure times.

The dataset spans multiple years, covering different evolutionary stages of the remnant. A large number of ObsIDs were analyzed to ensure comprehensive temporal coverage. Due to the extensive nature of the dataset, a representative subset is discussed in the main analysis, while the complete list of ObsID is provided in Appendix A.

The use of multi-epoch data enables the study of long-term trends in X-ray flux and spectral evolution [1], [4].

3.3 Data Reduction Procedures

The raw observational data were processed using standard data reduction procedures. This involved cleaning the event files, removing background noise, and applying calibration corrections. The primary steps included:

- Filtering of high background intervals to ensure data quality
- Application of calibration files for instrument response correction
- Extraction of cleaned event files for further analysis

These steps ensure that the extracted spectra accurately represent the physical properties of the source[11].

3.4 Spectral Extraction (Source & Background)

Spectral extraction was performed by defining appropriate source and background regions on the detector. The source region was selected to encompass the emission from SN 1987A, while the background region was chosen from a nearby source-free area to account for instrumental and cosmic background contributions[11].

The extracted spectra consist of photon counts as a function of energy, typically represented in units of counts $\text{s}^{-1} \text{keV}^{-1}$. Background subtraction was performed to isolate the intrinsic emission from the source.

A representative example of source and background extraction regions is shown in Figure 3.1.

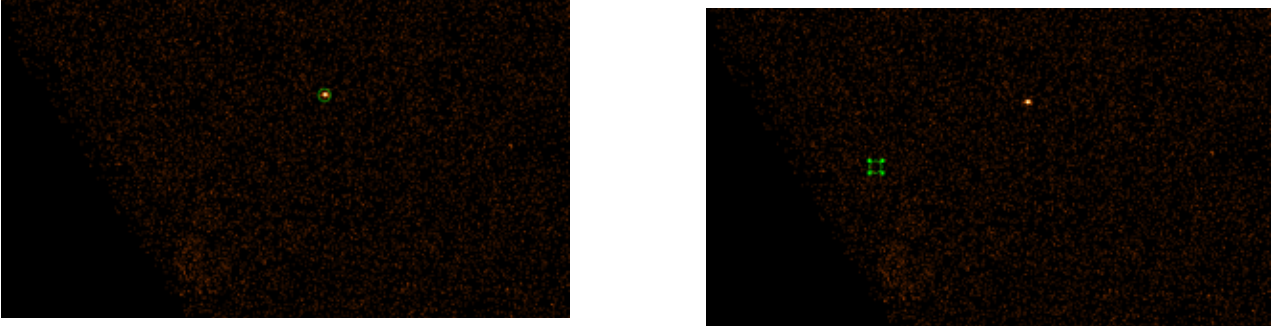


Figure 3.1: Source and background extraction regions for SN 1987A for Chandra observation (ObsID 122).

As shown in Figure 3.1, the source region encloses the emission from SN 1987A, while the background region is selected from a nearby source-free region to ensure accurate subtraction of background contributions.

3.5 Energy Band Selection (Soft & Hard X-ray)

To study the spectral evolution of SN 1987A, the X-ray emission was divided into two energy bands:

Soft X-ray band: 0.5-2 keV

Hard X-ray band: 3-8 keV

The soft X-ray band primarily traces emission from relatively cooler and dense plasma, typically associated with shock interaction with the equatorial ring. In contrast, the hard X-ray band corresponds to higher temperature plasma and is indicative of emission from less dense regions

and reverse-shocked ejecta[8].

This separation allows for a comparative analysis of different physical components contributing to the overall emission.

3.6 Flux Calculation Method

The X-ray flux was calculated from the extracted spectra using appropriate spectral models and instrument response files. The observed count rates were converted into physical flux units ($\text{erg cm}^{-2} \text{s}^{-1}$) by accounting for detector efficiency and effective area[12].

Unabsorbed flux values were determined to remove the effects of interstellar absorption, ensuring that the intrinsic emission properties of the source were accurately represented[12].

The flux values were calculated separately for soft and hard energy bands for each observation, enabling the construction of temporal evolution plots.

3.7 Hardness Ratio Definition

The hardness ratio (R) is defined as the ratio of hard X-ray flux to soft X-ray flux:

$$R = \frac{F_{hard}}{F_{soft}}$$

This parameter provides a measure of the relative contribution of high-energy emission compared to low-energy emission. Variations in the hardness ratio reflect changes in plasma temperature, density, and emission mechanisms[8].

The hardness ratio is a key diagnostic tool used in this study to identify different evolutionary phases of SN 1987A.

3.8 Error Analysis

Uncertainties in the flux measurements arise from statistical fluctuations in photon counts, background subtraction, and instrumental effects[11]. Error bars were estimated using standard statistical methods, typically based on Poisson uncertainties in the observed counts.

These uncertainties were propagated through the flux calculation and hardness ratio estimation. The inclusion of error analysis ensures the reliability and robustness of the derived trends and

interpretations[12].

Overall, the methodology adopted in this study provides a consistent and systematic framework for analyzing the long-term X-ray evolution of SN 1987A.

CHAPTER - 4

SPECTRAL ANALYSIS

4.1 Representative X-ray Spectrum

The X-ray spectrum of SN 1987A provides crucial information about the physical conditions of the emitting plasma. A representative spectrum was extracted from one of the observations (ObsID 122) to illustrate the general spectral characteristics.

The spectrum consists of photon counts as a function of energy and shows a typical thermal emission profile, with higher counts at lower energies and a gradual decline toward higher energies. This behaviour is consistent with emission from shock-heated plasma in the supernova remnant[8].

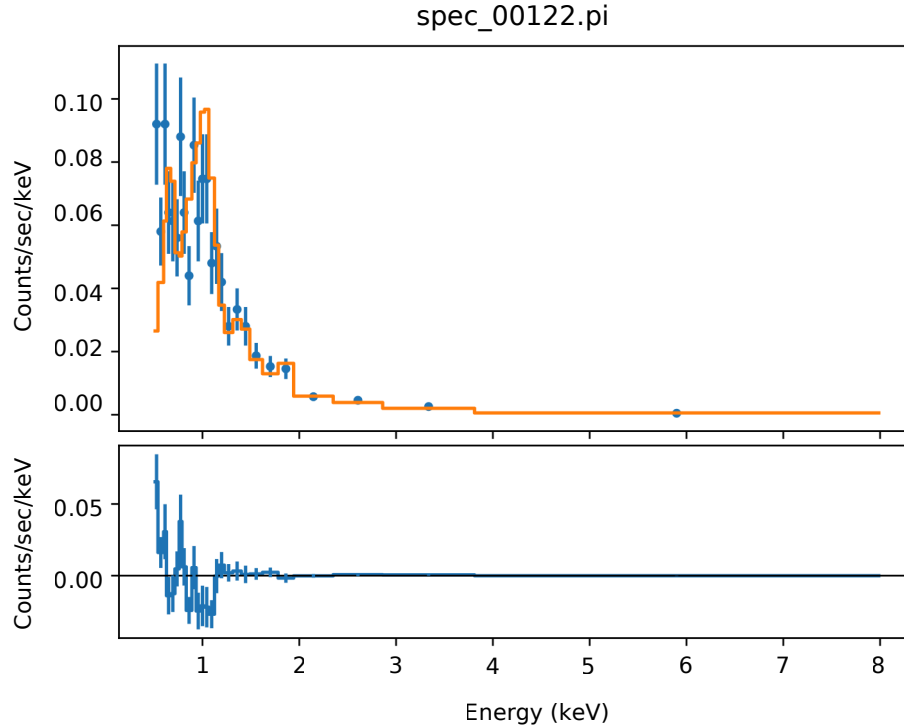


Figure 4.1: Representative X-ray spectrum with best-fit thermal plasma model and residuals (ObsID 122)

As shown in Figure 4.1, the spectrum exhibits strong emission in the soft X-ray region, indicating the presence of relatively cooler and denser plasma associated with the interaction of the shock wave with the equatorial ring[3].

4.2 Model fitting (Thermal Plasma Models)

To interpret the observed spectra, thermal plasma emission models were employed. These models

account for the emission processes occurring in hot, ionized gas, including thermal bremsstrahlung and line emission from highly ionized elements[9].

The fitting procedure involves comparing the observed spectrum with a theoretical model and adjusting parameters such as plasma temperature, emission measure, and elemental abundance to achieve the best fit. The model curve shown in Figure 4.1 closely follows the observed data, indicating that the emission is predominantly thermal in nature[8].

The presence of emission lines further supports the interpretation of a hot, ionized plasma environment within the remnant.

4.3 Residual Analysis

Residual analysis provides a measure of the difference between the observed data and the fitted model. It is an important diagnostic tool for evaluating the quality of the spectral fit.

In Figure 4.1, the residuals are plotted in the lower panel and represent the deviations of the observed counts from the model prediction. The residuals are generally distributed around zero, indicating a reasonably good fit between the model and the data[12].

Small deviations were observed at certain energy ranges, which may arise from uncertainties in the model parameters, instrumental effects, or the presence of multiple plasma components that are not fully captured by a single-temperature model[2].

4.4 Spectral Evolution Across Epochs

The spectral properties of SN 1987A evolve as the shock wave interacts with different regions of the circumstellar medium[7]. Analysis of spectra from multiple ObsIDs reveals changes in both flux levels and spectral shape. During the ring-dominated phase, the spectrum is characterized by strong soft X-ray emission due to interaction with dense material[3]. As the remnant evolves and the shock propagates beyond the equatorial ring, the contribution from higher energy emission becomes more significant[6].

This evolution reflects changes in plasma temperature and density, as well as the increasing influence of reverse-shocked ejecta[8]. The gradual shift toward higher energy emissions is consistent with the trends observed in the hardness ratio and flux evolution discussed in the subsequent chapter[4].

Overall, the spectral analysis confirms that the X-ray emission from Sn 1987A is governed by shock-heated thermal plasma and that its properties evolve in response to the changing interaction between the shock wave and the surrounding medium[7].

CHAPTER - 5

RESULTS AND DISCUSSION

5.1 Soft X-ray Flux Evolution (0.5-2 keV)

The evolution of the soft X-ray flux of SN 1987A, in the energy range 0.5-2 keV, is shown in Figure 5.1. The soft X-ray emission primarily traces relatively cooler and denser plasma associated with the interaction of the forward shock with the circumstellar material.

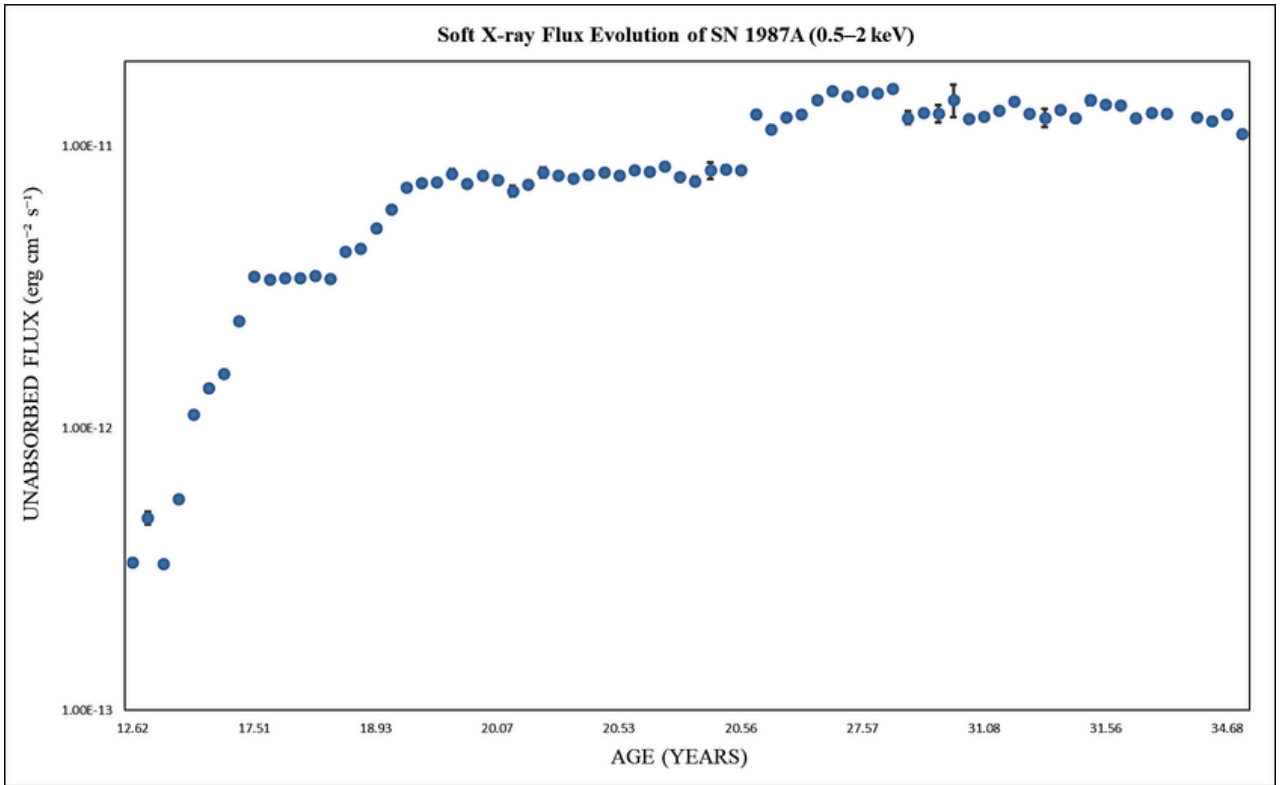


Figure 5.1: Soft X-ray flux evolution (0.5–2 keV) of SN 1987A as a function of age

As shown in Figure 5.1, the soft X-ray flux increases rapidly from ~13 to ~18 years, indicating the initial interaction of the shock with denser regions of the circumstellar medium. Between ~18 and ~25 years, the flux continues to rise but at a slower rate, corresponding to the ring-dominated phase where the shock interacts with the dense equatorial ring.

Beyond ~25 years, the soft X-ray flux shows a clear flattening and slight decline. This behaviour suggests that the forward shock has traversed the densest portion of the equatorial ring and is now propagating into lower-density material[3],[5].

5.2 Hard X-ray Flux Evolution

The hard X-ray flux evolution, in the energy range 3–8 keV, is presented in Figure 5.2. These components have higher temperature plasma and energetic processes within the remnant.

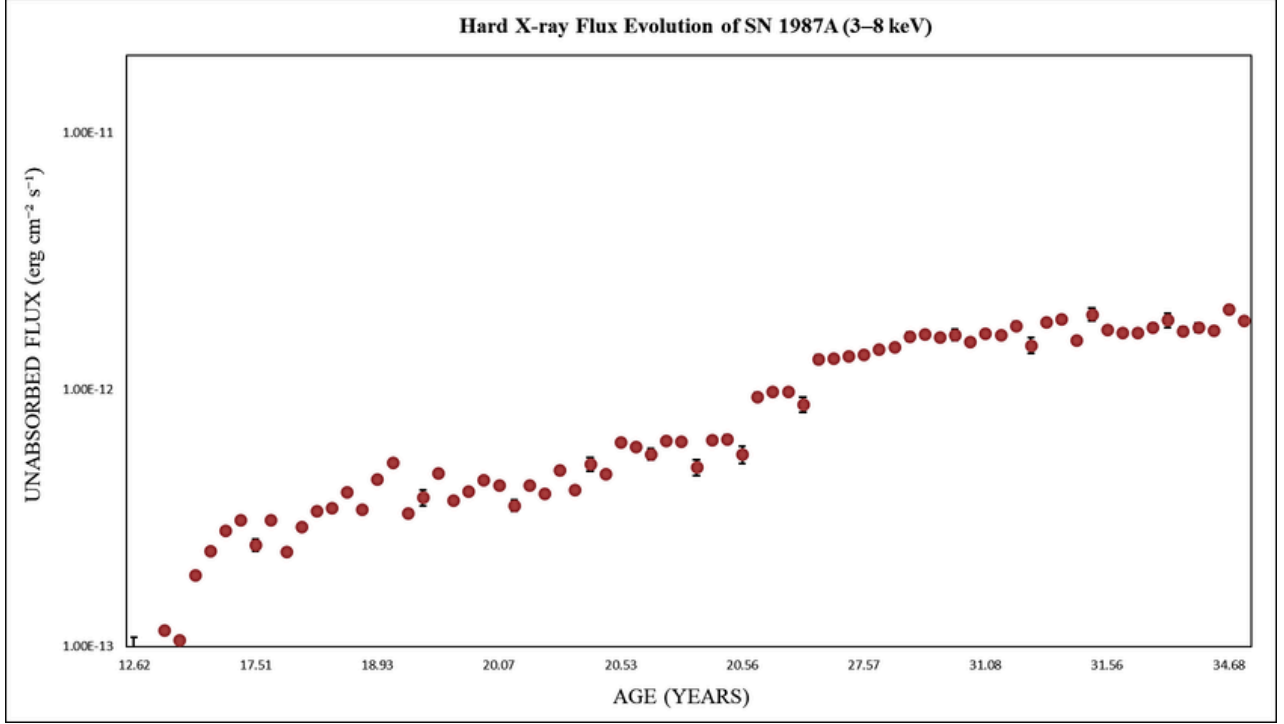


Figure 5.2: Hard X-ray flux evolution (3–8 keV) of SN 1987A as a function of age

Unlike the soft X-ray flux, the hard X-ray emission shows a gradual and continuous increase over the entire time range. The increase is relatively modest during the early phase but becomes more pronounced after ~25 years.

The sustained rise in hard X-ray flux indicates an increasing contribution from higher temperature plasma, likely associated with reverse-shocked ejecta and lower-density regions. This trend suggests that energetic processes become more dominant as the remnant evolves[6],[8].

5.3 Hardness Ratio Evolution

The hardness ratio, defined as the ratio of hard to soft X-ray flux, provides a measure of the relative contribution of high-energy emission.

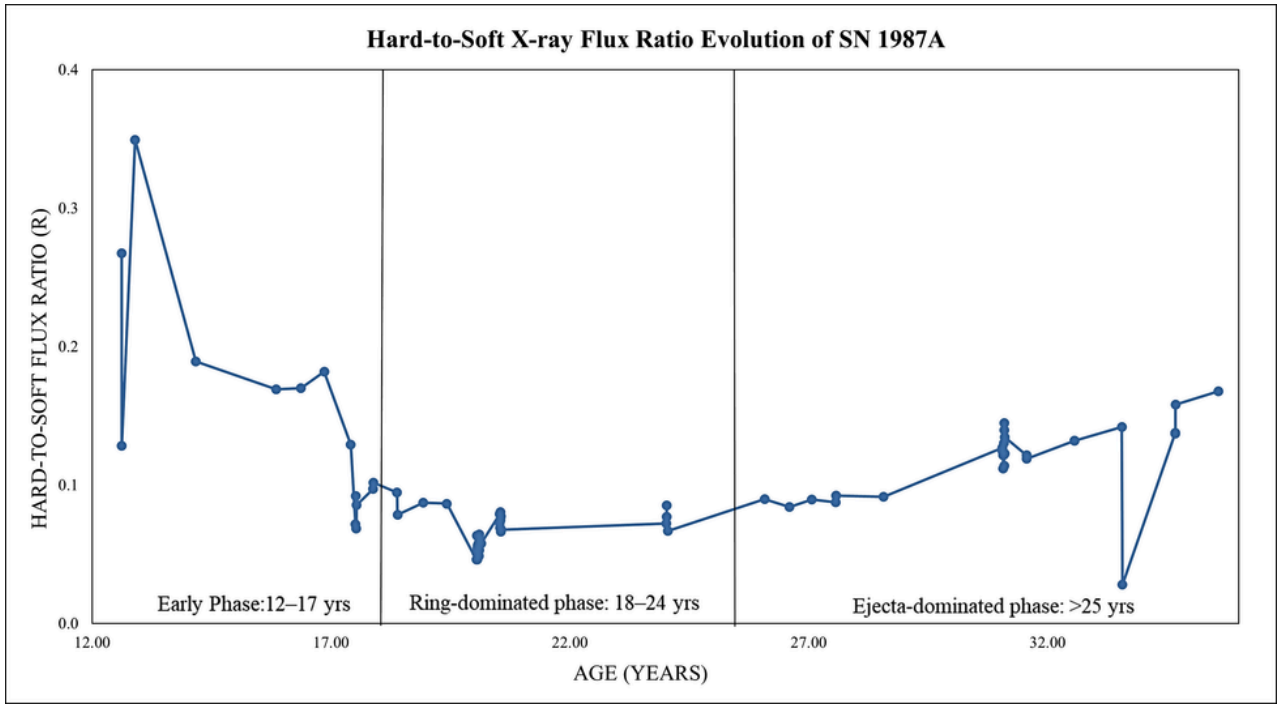


Figure 5.3: Hardness ratio evolution of SN 1987A

As shown in Figure 5.3, the hardness ratio is relatively high during the early phase (~12-17 years), indicating a comparatively stronger contribution from hard X-ray emission when the overall flux is low. During the ring-dominated phase (~18-24 years), the hardness ratio decreases significantly, reflecting the dominance of soft X-ray emission from interactions with the dense equatorial ring.

After ~25 years, the hardness ratio begins to increase steadily. This trend indicates a shift toward higher energy emissions and suggests that the remnant is transitioning into a new evolutionary phase dominated by hotter plasma components[6].

5.4 Phase Classification of SN 1987A

Based on the observed trends in flux and hardness ratio, the evolution of SN 1987A can be classified into three distinct phases:

- **Early Phase (~12-17 years):** Characterized by low flux levels and relatively higher hardness ratios due to emission from a low-density medium.
- **Ring-Dominated Phase (~18-24 years):** Marked by a strong increase in soft X-ray flux and a decrease in hardness ratio, indicating dominant emission from dense circumstellar material.
- **Ejecta-Dominated Phase(>25):** Characterized by flattening of the soft X-ray flux and increasing hard X-ray emission, reflecting contributions from reverse-shocked ejecta and

higher temperature plasma.

This classification is consistent with the physical evolution of the shock interaction[2],[3].

Table 5.1: Phase-wise Classification of SN 1987A Evolution

Phase	Age (years)	Dominant Emission	Physical Meaning
Early Phase	12–17	Soft X-ray dominant (moderate hardness ratio)	Shock in low-density medium
Ring-Dominated Phase	18–24	Soft X-ray dominant	Interaction with dense ring
Ejecta-Dominated Phase	>25	Hard X-ray increasing	Reverse shock heating ejecta

5.5 Linear Fit and Statistical Analysis

To quantify the late-time trend in the hardness ratio, a linear fit was applied to the post-transition data.

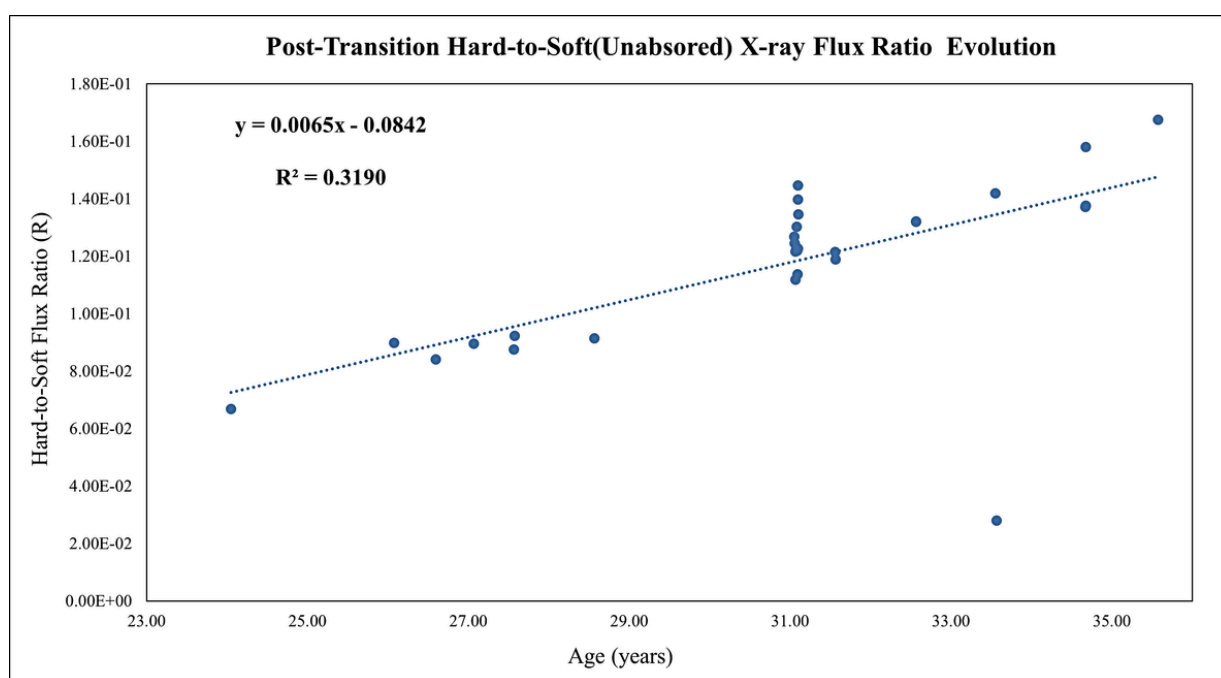


Figure 5.4: Linear regression of late-time hardness ratio as a function of age

The fitted relation shows a positive slope, indicating a steady increase in the hardness ratio over time. The coefficient of determination ($R^2 \sim 0.32$) suggests a moderate correlation, implying that while the overall trend is increasing, there is significant scatter in the data.

This scatter reflects the complex nature of the circumstellar medium and the presence of inhomogeneities in the emitting regions.

5.6 Residual Analysis of Late-Time Data

Residual analysis was performed to evaluate the quality of the linear fit.

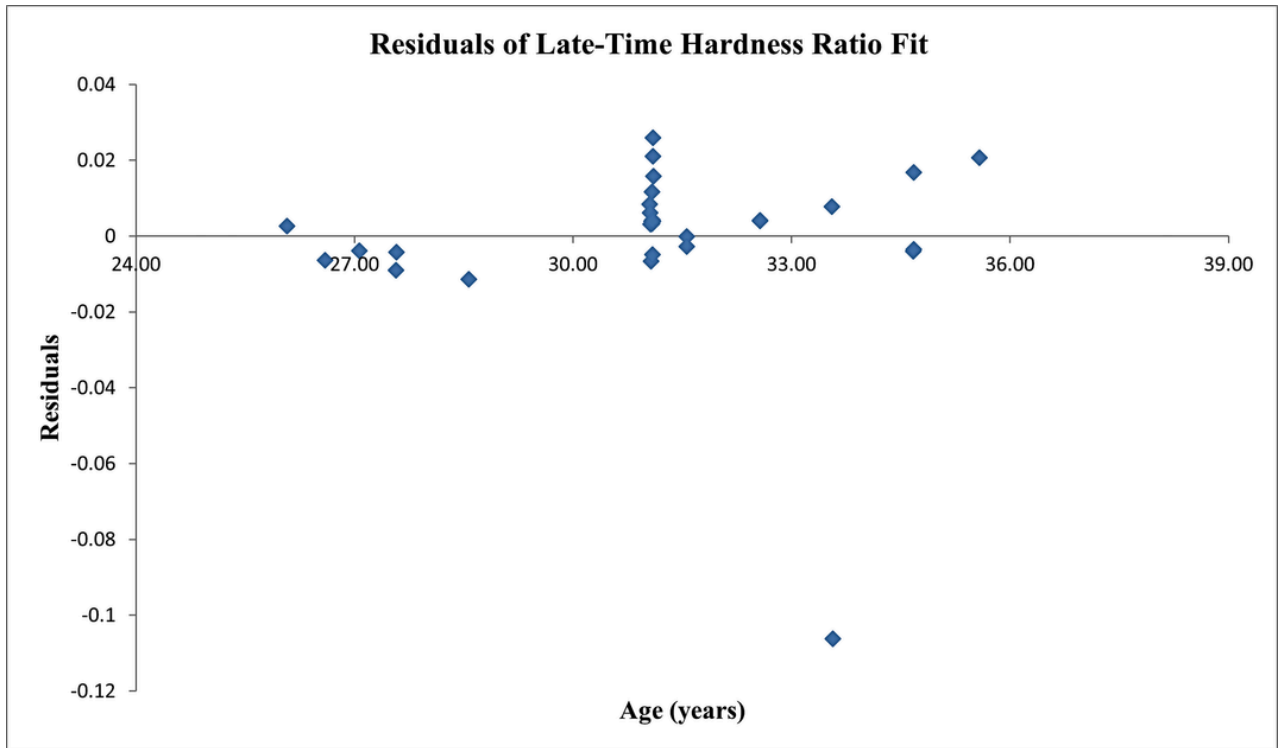


Figure 5.5: Residuals of the linear fit to late-time hardness ratio

As shown in Figure 5.5, the residuals are distributed around zero, indicating that the linear model provides a reasonable approximation of the trend. However, noticeable deviations are present, particularly at certain epochs, suggesting that additional physical processes or variations in local density may influence the emission.

5.7 Physical interpretation of Shock Evolution

The observed evolution of X-ray flux can be interpreted in terms of shock dynamics and plasma

properties. The strong increase in soft X-ray flux during the intermediate phase confirms that the emission is dominated by dense regions of the circumstellar medium[3], [8], consistent with the dependence of X-ray emission on the square of the plasma density[9].

As the shock propagates beyond the equatorial ring, the density decreases, leading to a reduction in soft X-ray emission. Simultaneously, the increase in hard X-ray flux indicates the presence of higher temperature plasma and the growing importance of reverse-shocked ejecta[6].

This behavior reflects a transition from density-dominated emission to temperature-dominated emission, which is a key characteristic of evolving supernova remnants.

5.8 Transition to Ejecta-Dominated Phase

The combined analysis of soft and hard X-ray fluxes, along with the hardness ratio, provides strong evidence that SN 1987A has entered an ejecta-dominated phase[6].

The flattening of soft X-ray flux and the continued rise in hard X-ray emission indicate that the forward shock has moved beyond the dense equatorial ring and is now interacting with lower-density material. At the same time, the reverse shock is heating the ejecta, contributing significantly to high-energy emission.

This transition marks a critical stage in the evolution of the remnant and provides important insights into the long-term behaviour of supernova remnants.

Overall, the results demonstrate that the evolution of SN 1987A is governed by complex interactions between shock waves, plasma conditions, and the structure of the surrounding medium[7].

CHAPTER - 6

THEORETICAL FRAMEWORK

6.1 Shock Physics in Supernova Remnants

Supernova remnants evolve through the interaction of an expanding shock wave with the surrounding circumstellar medium (CSM)[1],[2]. When a massive star undergoes a supernova explosion, the ejected material expands outward at supersonic velocities, generating a strong forward shock. This shock compresses and heats the ambient medium, producing high-energy emission, particularly in the X-ray band.

In addition to the forward shock, a reverse shock propagates inward through the ejecta[2],[3]. This reverse shock heats the inner material and contributes significantly to the high-energy emission at later stages of evolution. The interaction between these shocks and the surrounding medium governs the dynamical and radiative evolution of the remnant.

The structure of the surrounding medium plays a crucial role in determining the observed emission[3],[5]. In the case of SN 1987A, the presence of a dense equatorial ring leads to enhanced shock interaction, resulting in strong X-ray emission during the intermediate phase.

6.2 Thermal Bremsstrahlung Emission

The primary mechanism responsible for X-ray emission in supernova remnants is thermal bremsstrahlung, or free-free emission[9]. This process occurs when high-energy electrons are decelerated in the electric field of ions, in the emission of radiation.

The emissivity of thermal bremsstrahlung is given by:

$$eff \propto n_e^2 T^{\frac{1}{2}}$$

Where n_e is the electron density, and T is the plasma temperature.

This relation indicates that the emitted radiation depends strongly on both the density and temperature of the plasma. At the high temperature (10^6 - 10^7 K) typical of supernova remnants, thermal bremsstrahlung produces significant X-ray emission[10].

In addition to continuum emission, line emission from highly ionized elements such as oxygen, neon, silicon, and iron contributes to the observed spectra, confirming the thermal nature of the plasma.

6.3 Density Dependence of X-ray Flux

The observed X-ray flux[9] from a supernova remnant depends on the density of the emitting plasma. The total flux can be expressed as

$$\varepsilon \propto n_e^2 V$$

Where n_e is the electron density, and V is the emitting volume.

This quadratic dependence on density implies that regions with higher density produce significantly stronger X-ray emission[3], [9]. In SN 1987A, the dense equatorial ring provides such a region, leading to the observed increase in soft X-ray flux during the ring-dominated phase.

As the shock wave propagates beyond the ring into lower-density regions, the X-ray flux decreases accordingly. This behaviour is consistent with the trends observed in the soft X-ray flux evolution.

6.4 Temperature and Shock Velocity Relations

The temperature of the shock-heated plasma is directly related to the velocity of the shock wave. For a strong adiabatic shock[1],[2], the post-shock temperature is given by:

$$T \propto v_s^2$$

Where μ is the mean molecular weight, m_p is the proton mass, K is the Boltzmann constant, and v_s is the shock velocity.

This relation shows that higher shock velocities result in higher plasma temperatures. As the remnant evolves, variations in shock velocity and ambient density lead to changes in plasma temperature and, consequently, the observed X-ray emission.

In the later stages of SN 1987A[6],[8], the increasing contribution of hard X-ray emission suggests the presence of higher temperature plasma, consistent with the influence of reverse shocks and evolving shock dynamics.

CHAPTER - 7

CONCLUSION AND FUTURE SCOPE

This study presents a comprehensive analysis of the long-term X-ray evolution of the supernova remnant SN 1987A using multi-epoch observational data. The temporal behaviour of soft (0.5-2 keV) and hard (3-8 keV) X-ray fluxes, along with the hardness ratio, was examined to understand the interaction between the expanding shock wave and the surrounding circumstellar medium.

The results show that the soft X-ray flux increased significantly during the early and intermediate stages, corresponding to the interaction of the forward shock with the dense equatorial ring. This phase was characterized by efficient shock heating of high-density material, leading to enhanced emission. At later times, the soft X-ray flux exhibited a clear flattening and slight decline, indicating that the shock has propagated beyond the densest regions of the ring.

In contrast, the hard X-ray flux displayed a steady and continuous increase over time. This behaviour suggests an increasing contribution from higher temperature plasma and reverse-shocked ejecta. The gradual dominance of hard X-ray emission reflects the evolving physical conditions within the remnant.

The hardness ratio analysis provided strong evidence for a transition in the dominant emission processes. The decrease in the hardness ratio during the ring-dominated phase confirms the dominance of soft X-ray emission, while the subsequent increase indicates a shift toward higher energy emission in the later stages. The linear fit to the late-time data revealed a positive trend with moderate correlation, highlighting the complex and non-uniform nature of the circumstellar medium.

Based on the observed trends, the evolution of SN 1987A was classified into three distinct phases: an early phase characterized by low emission, a ring-dominated phase with strong soft X-ray emission, and a later ejecta-dominated phase marked by increasing hard X-ray contribution. This classification is consistent with theoretical models of supernova remnant evolution.

The results demonstrate that X-ray emission from SN 1987A is governed by the interplay between shock dynamics, plasma temperature, and density variations. The transition from density-dominated emission to temperature-dominated emission represents a key stage in the evolution of the remnant. Overall, this study provides a clear observational characterization of the transition phase of SN 1987A and contributes to a deeper understanding of shock interactions and plasma processes in young supernova remnants.

Future work may involve detailed spectral modelling using multi-temperature plasma components and the incorporation of additional observational data to further refine the understanding of the remnant's evolution.

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APPENDICES

Table A.1: Basic Parameters for SN 1987A

ObsID	INSTRUMENT	kT (keV)	Counts (net)	Exposure Time (ks)	Model Used
124	ACIS-S	32.3139	323	47.33056324	xsphabs.abs1 * xsvpshock.plasma
1387	ACIS-S	10.5047	466	68.89188012	xsphabs.abs1 * xsvpshock.plasma
122	ACIS-S	12.7462	640	8.559883613	xsphabs.abs1 * xsvpshock.plasma
1044	ACIS-S	10.1247	1851	17.78216673	xsphabs.abs1 * xsvpshock.plasma
3829	ACIS-S	10.8988	9421	49.0135041	xsphabs.abs1 * xsvpshock.plasma
3830	ACIS-S	10.9996	9802	45.30905939	xsphabs.abs1 * xsvpshock.plasma
4614	ACIS-S	11.6247	12034	46.49323962	xsphabs.abs1 * xsvpshock.plasma
4615	ACIS-S	8.98217	18247	48.83342139	xsphabs.abs1 * xsvpshock.plasma
5362	ACIS-S	6.37134	3866	67.41618669	xsphabs.abs1 * xsvpshock.plasma
4641	ACIS-S	7.82309	4148	72.48614733	xsphabs.abs1 * xsvpshock.plasma
6099	ACIS-S	4.75638	2841	48.61243982	xsphabs.abs1 * xsvpshock.plasma
5363	ACIS-S	6.87671	2584	43.45350731	xsphabs.abs1 * xsvpshock.plasma
5579	ACIS-S	6.62796	16557	31.86882774	xsphabs.abs1 * xsvpshock.plasma
6178	ACIS-S	6.27168	8698	16.47622324	xsphabs.abs1 * xsvpshock.plasma
5580	ACIS-S	7.06933	14779	23.74933217	xsphabs.abs1 * xsvpshock.plasma
6345	ACIS-S	5.4624	13040	20.99217238	xsphabs.abs1 * xsvpshock.plasma
6668	ACIS-S	5.93463	31350	42.33832017	xsphabs.abs1 * xsvpshock.plasma
6669	ACIS-S	5.51298	31372	36.44599803	xsphabs.abs1 * xsvpshock.plasma
8523	ACIS-S	3.5421	2362	29.65215312	xsphabs.abs1 * xsvpshock.plasma
8537	ACIS-S	2.95162	1109	12.73234923	xsphabs.abs1 * xsvpshock.plasma
7588	ACIS-S	4.84722	2166	27.21219364	xsphabs.abs1 * xsvpshock.plasma

APPENDICES

Table A.1: Basic Parameters for SN 1987A

ObsID	INSTRUMENT	kT (keV)	Counts (net)	Exposure Time (ks)	Model Used
8538	ACIS-S	3.32808	1846	20.69477268	xsphabs.abs1 * xsvpshock.plasma
7589	ACIS-S	3.85759	2144	25.26973383	xsphabs.abs1 * xsvpshock.plasma
8539	ACIS-S	3.61579	2092	24.77725088	xsphabs.abs1 * xsvpshock.plasma
8542	ACIS-S	3.8563	1469	17.85732312	xsphabs.abs1 * xsvpshock.plasma
8487	ACIS-S	3.09273	2306	28.672164	xsphabs.abs1 * xsvpshock.plasma
8543	ACIS-S	3.57612	2620	30.6596507	xsphabs.abs1 * xsvpshock.plasma
8544	ACIS-S	3.37053	1672	19.11728136	xsphabs.abs1 * xsvpshock.plasma
8488	ACIS-S	4.52288	2753	31.66215105	xsphabs.abs1 * xsvpshock.plasma
8545	ACIS-S	4.55603	1784	20.46226496	xsphabs.abs1 * xsvpshock.plasma
8546	ACIS-S	4.1204	2748	30.64467118	xsphabs.abs1 * xsvpshock.plasma
7590	ACIS-S	4.19865	3061	35.55210389	xsphabs.abs1 * xsvpshock.plasma
9581	ACIS-S	5.1654	5994	44.96267762	xsphabs.abs1 * xsvpshock.plasma
9582	ACIS-S	4.18014	5951	44.1754934	xsphabs.abs1 * xsvpshock.plasma
9580	ACIS-S	4.05945	4570	34.59481094	xsphabs.abs1 * xsvpshock.plasma
7620	ACIS-S	4.31894	4750	34.62682547	xsphabs.abs1 * xsvpshock.plasma
7621	ACIS-S	4.33889	4829	36.953183	xsphabs.abs1 * xsvpshock.plasma
9591	ACIS-S	3.57485	1674	12.85107516	xsphabs.abs1 * xsvpshock.plasma
9592	ACIS-S	4.32267	1732	12.87665437	xsphabs.abs1 * xsvpshock.plasma
9589	ACIS-S	4.50179	5266	39.53240408	xsphabs.abs1 * xsvpshock.plasma
9590	ACIS-S	4.57494	3373	24.66213851	xsphabs.abs1 * xsvpshock.plasma
12145	ACIS-S	3.41494	7693	51.20691526	xsphabs.abs1 * xsvpshock.plasma

APPENDICES

Table A.1: Basic Parameters for SN 1987A

ObsID	INSTRUMENT	kT (keV)	Counts (net)	Exposure Time (ks)	Model Used
13238	ACIS-S	3.34236	8166	54.15937719	xsphabs.abs1 * xsvpshock.plasma
13239	ACIS-S	3.32189	7272	46.73695277	xsphabs.abs1 * xsvpshock.plasma
12146	ACIS-S	3.32192	4095	25.62483515	xsphabs.abs1 * xsvpshock.plasma
14697	ACIS-S	3.44652	11926	67.57127898	xsphabs.abs1 * xsvpshock.plasma
14698	ACIS-S	3.71004	12226	68.53495077	xsphabs.abs1 * xsvpshock.plasma
15809	ACIS-S	3.44652	12224	70.46209498	xsphabs.abs1 * xsvpshock.plasma
17415	ACIS-S	3.54326	3249	19.36873763	xsphabs.abs1 * xsvpshock.plasma
15810	ACIS-S	3.77901	8258	48.29073174	xsphabs.abs1 * xsvpshock.plasma
16756	ACIS-S	3.60893	10726	66.59796817	xsphabs.abs1 * xsvpshock.plasma
20927	ACIS-S	3.44657	2082	16.63956304	xsphabs.abs1 * xsvpshock.plasma
21037	ACIS-S	3.32184	3914	29.41814519	xsphabs.abs1 * xsvpshock.plasma
21038	ACIS-S	3.29138	4366	34.21905689	xsphabs.abs1 * xsvpshock.plasma
20322	ACIS-S	3.4149	1893	15.1571905	xsphabs.abs1 * xsvpshock.plasma
21042	ACIS-S	3.08588	5393	41.05285084	xsphabs.abs1 * xsvpshock.plasma
21043	ACIS-S	3.67602	3712	28.82843567	xsphabs.abs1 * xsvpshock.plasma
21044	ACIS-S	3.44656	1963	15.17912887	xsphabs.abs1 * xsvpshock.plasma
20323	ACIS-S	4.1436	3431	27.09621603	xsphabs.abs1 * xsvpshock.plasma
21049	ACIS-S	3.2914	3980	30.89900836	xsphabs.abs1 * xsvpshock.plasma
21050	ACIS-S	3.92083	2247	17.764949	xsphabs.abs1 * xsvpshock.plasma
21051	ACIS-S	3.81398	1983	15.00922042	xsphabs.abs1 * xsvpshock.plasma
21052	ACIS-S	3.23131	3778	30.11543946	xsphabs.abs1 * xsvpshock.plasma

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Table A.1: Basic Parameters for SN 1987A

ObsID	INSTRUMENT	kT (keV)	Counts (net)	Exposure Time (ks)	Model Used
21053	ACIS-S	4.06798	1621	12.18729925	xsphabs.abs1 * xsvpshock.plasma
20277	ACIS-S	3.26121	4541	33.82601368	xsphabs.abs1 * xsvpshock.plasma
21844	ACIS-S	3.38373	4517	33.82811733	xsphabs.abs1 * xsvpshock.plasma
21304	ACIS-S	3.29141	4933	41.05631531	xsphabs.abs1 * xsvpshock.plasma
22849	ACIS-S	3.67605	4950	41.05630622	xsphabs.abs1 * xsvpshock.plasma
22425	ACIS-S	3.81398	7020	60.8205053	xsphabs.abs1 * xsvpshock.plasma
24652	ACIS-S	1.73804	3418	29.33136635	xsphabs.abs1 * xsvpshock.plasma
23534	ACIS-S	3.17233	2984	27.07967686	xsphabs.abs1 * xsvpshock.plasma
24295	ACIS-S	3.43018	3115	29.00681893	xsphabs.abs1 * xsvpshock.plasma
24654	ACIS-S	3.74436	3403	31.89969408	xsphabs.abs1 * xsvpshock.plasma
25514	ACIS-S	2.97426	5972	56.9643148	xsphabs.abs1 * xsvpshock.plasma

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Table A.2: Flux Parameters for SN 1987A

ObsID	Epoch (years)	Soft Flux (un-abs) (erg cm⁻² s⁻¹)	Hard Flux (un-abs) (erg cm⁻² s⁻¹)	R (RATIO) (Unabs)
124	12.61642	3.35E-13	8.96E-14	2.67E-01
1387	12.61803	4.81E-13	6.18E-14	1.28E-01
122	12.8998	3.30E-13	1.15E-13	3.49E-01
1044	14.1703	5.62E-13	1.06E-13	1.89E-01
3829	15.85251	1.12E-12	1.90E-13	1.69E-01
3830	16.37015	1.39E-12	2.36E-13	1.70E-01
4614	16.85883	1.56E-12	2.84E-13	1.82E-01
4615	17.4103	2.41E-12	3.12E-13	1.29E-01
5362	17.51038	3.47E-12	2.48E-13	7.16E-02
4641	17.51772	3.38E-12	3.12E-13	9.21E-02
6099	17.52489	3.41E-12	2.34E-13	6.85E-02
5363	17.53324	3.41E-12	2.92E-13	8.56E-02
5579	17.87962	3.48E-12	3.38E-13	9.70E-02
6178	17.89096	3.40E-12	3.46E-13	1.02E-01
5580	18.38162	4.24E-12	4.01E-13	9.46E-02
6345	18.39375	4.35E-12	3.41E-13	7.85E-02
6668	18.93017	5.14E-12	4.48E-13	8.72E-02
6669	19.42477	6.00E-12	5.19E-13	8.65E-02
8523	20.04425	7.15E-12	3.31E-13	4.62E-02
8537	20.04865	7.45E-12	3.80E-13	5.10E-02

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Table A.2: Flux Parameters for SN 1987A

ObsID	Epoch (years)	Soft Flux (un-abs) (erg cm⁻² s⁻¹)	Hard Flux (un-abs) (erg cm⁻² s⁻¹)	R (RATIO) (Unabs)
7588	20.05159	7.47E-12	4.74E-13	6.34E-02
8538	20.06338	8.02E-12	3.71E-13	4.62E-02
7589	20.06592	7.40E-12	4.02E-13	5.44E-02
8539	20.06988	7.89E-12	4.45E-13	5.64E-02
8542	20.07332	7.62E-12	4.25E-13	5.58E-02
8487	20.08053	6.95E-12	3.54E-13	5.10E-02
8543	20.0877	7.35E-12	4.24E-13	5.77E-02
8544	20.09239	8.09E-12	3.95E-13	4.88E-02
8488	20.09505	7.89E-12	4.87E-13	6.18E-02
8545	20.09944	7.70E-12	4.09E-13	5.31E-02
8546	20.10223	7.95E-12	5.12E-13	6.44E-02
7590	20.14562	8.09E-12	4.69E-13	5.79E-02
9581	20.52939	7.93E-12	6.27E-13	7.91E-02
9582	20.53308	8.25E-12	5.98E-13	7.26E-02
9580	20.53734	8.14E-12	5.61E-13	6.90E-02
7620	20.54378	8.50E-12	6.33E-13	7.44E-02
7621	20.54779	7.80E-12	6.27E-13	8.04E-02
9591	20.55111	7.51E-12	4.98E-13	6.63E-02
9592	20.55291	8.23E-12	6.36E-13	7.72E-02
9589	20.55834	8.30E-12	6.41E-13	7.72E-02
9590	20.56206	8.26E-12	5.59E-13	6.77E-02

APPENDICES

Table A.2: Flux Parameters for SN 1987A

ObsID	Epoch (years)	Soft Flux (un-abs) (erg cm⁻² s⁻¹)	Hard Flux (un-abs) (erg cm⁻² s⁻¹)	R (RATIO) (Unabs)
12145	24.01862	1.30E-11	9.37E-13	7.21E-02
13238	24.02592	1.15E-11	9.82E-13	8.52E-02
13239	24.03028	1.27E-11	9.80E-13	7.71E-02
12146	24.0513	1.31E-11	8.73E-13	6.69E-02
14697	26.07487	1.46E-11	1.31E-12	8.98E-02
14698	26.59611	1.57E-11	1.32E-12	8.40E-02
15809	27.06705	1.51E-11	1.36E-12	8.95E-02
17415	27.56709	1.56E-11	1.37E-12	8.75E-02
15810	27.57388	1.55E-11	1.43E-12	9.23E-02
16756	28.56604	1.60E-11	1.46E-12	9.14E-02
20927	31.05271	1.27E-11	1.61E-12	1.27E-01
21037	31.05659	1.32E-11	1.64E-12	1.25E-01
21038	31.06637	1.31E-11	1.59E-12	1.22E-01
20322	31.06886	1.46E-11	1.64E-12	1.12E-01
21042	31.07832	1.25E-11	1.54E-12	1.22E-01
21043	31.08284	1.27E-11	1.66E-12	1.30E-01
21044	31.08618	1.34E-11	1.64E-12	1.22E-01
20323	31.09021	1.44E-11	1.78E-12	1.23E-01
21049	31.09276	1.31E-11	1.49E-12	1.14E-01
21050	31.09566	1.27E-11	1.83E-12	1.45E-01

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Table A.2: Flux Parameters for SN 1987A

ObsID	Epoch (years)	Soft Flux (un-abs) (erg cm⁻² s⁻¹)	Hard Flux (un-abs) (erg cm⁻² s⁻¹)	R (RATIO) (Unabs)
21051	31.09784	1.35E-11	1.89E-12	1.40E-01
21052	31.10005	1.27E-11	1.55E-12	1.23E-01
21053	31.1051	1.46E-11	1.96E-12	1.35E-01
20277	31.56055	1.41E-11	1.72E-12	1.22E-01
21844	31.56354	1.40E-11	1.67E-12	1.19E-01
21304	32.56447	1.26E-11	1.66E-12	1.32E-01
22849	32.56699	1.32E-11	1.74E-12	1.32E-01
22425	33.55457	1.31E-11	1.86E-12	1.42E-01
24652	33.56632	6.05E-11	1.69E-12	2.80E-02
23534	34.67175	1.27E-11	1.74E-12	1.37E-01
24295	34.67666	1.23E-11	1.70E-12	1.38E-01
24654	34.67809	1.30E-11	2.05E-12	1.58E-01
25514	35.57603	1.11E-11	1.86E-12	1.68E-01

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Table A.3: Uncertainties in Flux Measurements for SN 1987A

ObsID	Unabsorbed Hard uncertainty	Unabsorbed Soft uncertainty	σ_R (Unabs)
124	1.93E-14	4.09E-15	5.77E-02
1387	7.42E-15	2.58E-14	1.69E-02
122	3.34E-16	2.75E-15	3.08E-03
1044	1.94E-15	1.45E-14	5.98E-03
3829	5.32E-16	8.83E-15	1.41E-03
3830	8.48E-16	3.27E-15	7.29E-04
4614	2.40E-16	5.09E-15	6.13E-04
4615	1.25E-15	9.68E-15	7.33E-04
5362	1.29E-14	3.89E-14	3.80E-03
4641	7.89E-15	4.52E-14	2.64E-03
6099	4.07E-15	4.87E-14	1.54E-03
5363	8.21E-15	7.61E-14	3.07E-03
5579	7.44E-15	9.09E-15	2.15E-03
6178	1.01E-14	3.53E-14	3.15E-03
5580	4.59E-16	8.62E-15	2.21E-04
6345	8.41E-15	2.91E-14	2.01E-03
6668	2.13E-15	2.45E-14	5.87E-04
6669	2.05E-15	2.37E-14	4.83E-04
8523	1.03E-14	6.18E-14	1.50E-03
8537	2.65E-14	1.90E-13	3.79E-03
7588	6.00E-15	1.39E-13	1.43E-03

APPENDICES

Table A.3: Uncertainties in Flux Measurements for SN 1987A

ObsID	Unabsorbed Hard uncertainty	Unabsorbed Soft uncertainty	σ_R (Unabs)
8538	8.91E-15	2.95E-13	2.03E-03
7589	1.28E-14	4.93E-14	1.77E-03
8539	5.00E-15	2.00E-13	1.56E-03
8542	8.80E-16	2.65E-13	1.94E-03
8487	1.71E-14	3.01E-13	3.31E-03
8543	1.02E-16	1.08E-13	8.44E-04
8544	1.15E-16	3.44E-13	2.07E-03
8488	6.16E-15	1.63E-13	1.50E-03
8545	2.68E-15	5.38E-14	5.09E-04
8546	3.02E-14	9.93E-14	3.89E-03
7590	9.63E-15	1.20E-13	1.47E-03
9581	1.54E-14	1.01E-13	2.19E-03
9582	2.20E-14	1.40E-13	2.94E-03
9580	2.83E-14	5.10E-14	3.50E-03
7620	2.67E-15	1.01E-14	3.26E-04
7621	3.80E-15	4.59E-14	6.80E-04
9591	3.49E-14	2.77E-13	5.25E-03
9592	3.22E-15	5.73E-13	5.39E-03
9589	2.30E-14	2.68E-13	3.73E-03
9590	4.25E-14	1.05E-13	5.21E-03
12145	3.66E-14	1.13E-14	2.81E-03

APPENDICES

Table A.3: Uncertainties in Flux Measurements for SN 1987A

ObsID	Unabsorbed Hard uncertainty	Unabsorbed Soft uncertainty	σ_R (Unabs)
13238	2.62E-14	7.06E-14	2.33E-03
13239	6.62E-15	8.24E-14	7.21E-04
12146	6.00E-14	1.14E-13	4.63E-03
14697	5.13E-14	4.76E-14	3.52E-03
14698	1.55E-15	1.34E-13	7.23E-04
15809	1.79E-15	1.87E-14	1.62E-04
17415	4.43E-14	1.51E-13	2.96E-03
15810	3.18E-15	8.53E-14	5.48E-04
16756	3.77E-15	2.09E-14	2.64E-04
20927	6.70E-14	6.42E-13	8.32E-03
21037	1.43E-14	8.23E-14	1.34E-03
21038	2.12E-14	9.08E-13	8.58E-03
20322	8.43E-14	1.91E-12	1.57E-02
21042	2.15E-16	8.79E-14	8.58E-04
21043	3.38E-14	1.65E-13	3.14E-03
21044	3.17E-14	3.58E-14	2.38E-03
20323	6.41E-14	2.35E-16	4.43E-03
21049	1.09E-13	8.57E-14	8.37E-03
21050	4.27E-14	9.05E-13	1.35E-02
21051	5.41E-14	1.15E-13	4.19E-03
21052	3.04E-14	3.25E-13	5.31E-03

APPENDICES

Table A.3: Uncertainties in Flux Measurements for SN 1987A

ObsID	Unabsorbed Hard uncertainty	Unabsorbed Soft uncertainty	σ_R (Unabs)
21053	1.08E-13	5.19E-13	8.85E-03
20277	2.22E-14	2.13E-13	2.42E-03
21844	1.90E-14	1.13E-13	1.66E-03
21304	4.36E-15	7.87E-15	3.56E-04
22849	1.30E-14	8.57E-14	1.31E-03
22425	1.18E-13	1.43E-13	9.11E-03
24652	5.44E-14	4.29E-12	2.18E-03
23534	6.78E-14	1.03E-13	5.45E-03
24295	5.40E-14	3.05E-13	5.54E-03
24654	4.39E-14	3.78E-14	3.41E-03
25514	4.15E-14	3.55E-13	6.55E-03